SM Firoz Ahmed Fahim

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Research Interests

My research interests focus on LLM, Computer Vision, and QML, with an emphasis on multimodal intelligence, representation learning, efficient model architectures, and hybrid quantum-classical approaches to machine learning.

Education

05.2018 – 03.2023 **American International University Bangladesh**
Dhaka, Bangladesh B.Sc. in Computer Science and Engineering
CGPA: **3.45/4.00** (last 60 credits: **3.60/4.00**)
Thesis: *Suicidal Tendency Detection Using Machine Learning Algorithms* – Supervisor: Prof. Dr. Md. Asraf Ali

Research Experience

Detecting Consistent Hallucinations in Large Language Models
*Elite Research Lab – **SM Firoz Ahmed Fahim**, S. Hossain, A. Rahman* 2026
First author; accepted at IEEE BigData 2026

- Studied why hallucination detectors miss *consistently repeated* LLM errors, a failure case that sampling-based detectors such as Semantic Entropy cannot see, and compared 11 black-box detection signals on Qwen2.5-7B and Llama-3.1-8B.
- Evaluated on 1,117 questions (TruthfulQA, TriviaQA, SciQ) against seven baselines, including Semantic Entropy, SelfCheckGPT, P(True) and MetaQA; found that sampling-based detectors score near chance (AUROC 0.51–0.53), while a second model family reaches 0.65–0.69 at one sixth of MetaQA's cost.
- Pre-registered hypotheses; AUROC with bootstrap CIs, DeLong tests with Holm correction, ablations and human validation ($\kappa = 0.92$), in a fully automated, tested pipeline that generates every reported number from saved results.

Explicit 3D Geometry for AI-Generated Image Detection
Independent research – Supervisor: Raihan Uddin Ahmed 2026
SM Firoz Ahmed Fahim, Sole author; manuscript in preparation

- Designed a study testing whether 3D scene geometry (depth, surfaces, lighting, perspective) helps detect AI-generated images from unseen generators, by adding 26 geometry features to three leading detectors (B-Free, CO-SPY, UnivFD).
- Built a fair evaluation on 4 test sets with no overlap between training and test generators, using controls and rigorous statistics to check that any gain comes from geometry, not shortcuts.

Can a Quantum Kernel Pick Better Demonstrations? Few-Qubit Kernel Retrieval for Few-Shot LLM Classification
*Elite Research Lab – **SM Firoz Ahmed Fahim**, A. Rahman, S. Hossain* 2026
First author; manuscript in preparation Manuscript in preparation

- Tested whether quantum kernels (IQP, angle and projected feature maps in PennyLane) select better few-shot demonstrations for frozen LLMs than classical similarity, across 4 datasets, 17 selectors and 3 LLMs (Claude Haiku, Qwen2.5-1.5B/7B), including qubit-scaling, gate-noise and shot-noise studies.
- Proved that retrieval depends only on the ranking of similarities and introduced a rank-agreement check that predicts quantum–classical differences before any LLM is called.
- Pre-registered analysis with TOST equivalence tests, cluster bootstrap CIs, Holm correction and minimum detectable effects.

Publications

[1] "Advancing Sentiment Analysis Through Natural Language Processing and Deep Learning," *Accepted in 16th ICCCNT*, 2025.

[2] "Revolutionizing Academic Learning and Research with the Integration of Leap Motion and ARCore in Android," *Accepted in 16th ICCCNT*, 2025.

[3] "Healthcare Cost Patterns and Prediction: Investigating Personal Datasets using Data Analytics," M. A. Islam, A. Nag, P. Chandra, **SM Firoz Ahmed Fahim**, M. M. Hoque, *3rd International Conference on Signal and Data*

Processing, 2025. [link]

[4] "An Investigation into the Prediction of Annual Income Levels through the Utilization of Demographic Features Employing the Modified UCI Adult Dataset," M. A. Islam, A. Nag, N. Roy, A. R. Dey, **SM Firoz Ahmed Fahim**, A. Ghosh, *International Conference on Computing, Communication, and Intelligent Systems*, 2023. [link]

[5] "Utilization of Encoding, Early Stopping, Hyper Parameter Tuning, and Machine Learning Models for Bank Fraud Detection," M. A. Islam, A. Nag, S. Chowdhury, **SM Firoz Ahmed Fahim**, A. Ghosh, N. Mumtaj, *IEEE 9th International Women in Engineering Conference on Electrical and Computer Engineering*, 2023. [link]

[6] "A Survey on Wireless Sensor Network Routing Performance Optimizing and Security Techniques," A. Nag, D. Mandal, N. Roy, N. Chand, **SM Firoz Ahmed Fahim**, *4th International Conference on Frontiers in Computing and Systems (COMSYS)*, 2023. [link]

Work Experience

09.2026 – Present NY, USA
Research Intern, *Elite Research Lab LLC*
- Contribute to three research projects spanning large language models, quantum machine learning, and computer-assisted language learning, each targeting peer-reviewed publication.
- Built reproducible Python research pipelines with automated tests, cached results and scripts that generate every table, figure and reported number.

04.2025 – 12.2025 San Francisco, Remote
Web Scraper QA Specialist, *Reworkd (Y Combinator–backed, AI Grant–funded)*
- Built, reviewed and debugged automated scraping agents for structured and unstructured data, contributing to scalable high-volume scraping infrastructure.

07.2024 – 03.2025 Dhaka, Bangladesh
Research Assistant, *Artificial Intelligence Research & Innovation Lab (AIRIL)*
- Developed machine learning models for real-world prediction problems and analyzed data to improve them.

03.2021 – 04.2024 Dubai, Remote
Web Developer, *Fiume*
- Designed and implemented web applications and CMS-based systems with modern front-end technologies.

Teaching Experience

2022 Dhaka, Bangladesh
Undergraduate Teaching Assistant, *American International University Bangladesh*
Theory of Computation; Computer Graphics.

Selected Projects

Quantum CLIP – integrated quantum circuits and BERT into the CLIP architecture.

Smart Home Control via Hand Gestures – OpenCV system controlling lights and fans from camera-detected gestures.

Hydromint – SaaS platform for creating and managing NFT projects.

Web Series Management System – ASP.NET app with Admin, Streaming and Package Manager roles.

Skills

Programming Languages	Python, Java, ASP.NET
Machine Learning & NLP	Supervised learning, text classification, transformer-based models, sentence embeddings, information retrieval, in-context learning, model evaluation
Computer Vision	Image classification, image forensics, feature extraction, vision transformers, multimodal representation learning
Quantum Machine Learning	Quantum kernel methods, quantum feature maps, parameterized quantum circuits, circuit simulation
Frameworks & Libraries	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NumPy, SciPy, pandas, OpenCV, PennyLane
Statistical Analysis	Hypothesis testing, confidence intervals, bootstrap resampling, ROC/AUROC analysis, model comparison, cross-validation
Research Methods	Experimental design, ablation studies, baseline comparisons, error analysis, data preprocessing, reproducible experiments, literature review
Development Tools	Git/GitHub, Google Colab, Jupyter Notebook, LaTeX/BibTeX, REST APIs
Languages	Bangla (native), English (professional)

Leadership & Service

08.2026 – Present **President & Country Head**, *AIESEC in Bangladesh*

- Head of Bangladesh's national branch of the world's largest youth-run organisation, owning strategy, performance and execution across three business units and 250+ members.
- Direct a multinational executive board of six Vice Presidents from six countries, building a KPI-driven review and accountability system across all national functions.
- Serve as the organisation's chief representative to universities, NGOs, corporate partners and AIESEC's global network spanning 106 countries.

08.2025 – 07.2026 **National Head of Youth Operations**, *AIESEC in Bangladesh*

- Generated 1,000+ exchange sign-ups and 12 exchange approvals in one year, the national branch's best results in 8 years.
- Maintained 90% retention across a 42-person national team by introducing structured onboarding, coaching and OKR-based performance reviews.
- Built university partnerships across Dhaka and relationships with AIESEC entities in 20+ countries, and represented Bangladesh at the Asia Pacific conference in Nepal.

Awards

2026 **Zero to Hero Award**, *AIESEC - Asia Pacific Conference, Nepal*
Awarded to AIESEC in Bangladesh among 23 entities for growth in outgoing exchanges (oGX), a function I led as National Head of Youth Operations.

2019 **Best Presenter Award**, *American International University Bangladesh*

2016 **Government Scholarship**, *Jessore Education Board – 12th position in the board*

References

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